

Uzair Arif

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Skills

Languages & ML: Python, SQL, PyTorch, Scikit-learn, XGBoost, CatBoost, SciPy, NumPy, Pandas

LLMs & Agents: Multi-Agent Systems, LangGraph, Ollama, Model Context Protocol (MCP), LangChain, RAG (FAISS/BM25/RRF), Memory Architectures

MLOps & Tooling: Docker, FastAPI, MLflow, Pytest, Git, SQLite, Optuna, Streamlit

Core Concepts: Multi-Agent Simulation, Memory Collapse Dynamics, Hypothesis Testing (Mann-Whitney U, FDR), Controlled Experiment Design

Professional Experience

National Center of Artificial Intelligence (NCAI) — Research Intern

Jul 2025 – Aug 2025

- Conducted literature review on Multi-Agent RL, public goods games, and Ostrom's governance principles; drafted a full research proposal on emergent institutional design in cooperative agent systems.
- Designed core simulation components including democratic voting on punishment rules, weighted voting schemes, and IPPO agent architecture with reward functions for norm emergence.

Federal Board of Revenue (FBR) — Revenue Forecasting Intern

Jun 2025 – Jul 2025

- Built 5 daily Prophet models (Income Tax, Sales Tax, Federal Excise, Customs Duty, Gross Collection) with Pakistan-specific Gregorian and lunar holiday effects and monthly seasonality.
- Tuned all models via Optuna hyperparameter search and evaluated on held-out validation sets using MAE and MSE to handle highly volatile government revenue patterns.
- Automated a multi-horizon Excel reporting pipeline with daily, weekly, and monthly breakdowns; deployed an interactive Streamlit dashboard to surface mid-month surges and fiscal deadline spikes for the team.

National Aerospace Science & Technology Park (NASTP) — AI Intern

Jun 2024 – Aug 2024

- Engineered real-time data streaming and visualization tools using ZeroMQ and Python, reducing publish-subscribe latency to milliseconds.
- Built CNNs, linear regression, and SVM models from scratch, improving predictive accuracy by 10%.
- Created an anomaly detection model for ECG time-series data, achieving a 95% detection rate for anomalies.

Projects

SEAM — Shared Evolving Agent Memory

- Built a 120-run controlled study of self-evolving LLM agents (3 envs x 4 policies x 10 seeds). A playbook-based memory policy scored 4.2x above the no-memory baseline and beat a hand-coded oracle on 9 of 10 seeds (efficacy gap 1.53), significant at FDR $p = 0.0011$ (Cliff's delta = 1.0).
- Proved naive memory breaks agents: raw trajectory buffers closed zero deals across all 10 negotiation runs, while incremental memory reached a fair deal on every seed, showing curation, not memory presence, drives multi-agent performance.
- Built the metrics stack: Self-BLEU collapse detection, action entropy, oracle-normalized efficacy gap, Mann-Whitney U tests, and Benjamini-Hochberg FDR statistics.
- Engineered the platform end to end: 3 memory policies, JSONL logging with run rehydration, automated figures and reports, and 125 passing pytest tests.

ELICIT: Emergent Multi-Agent Climate Institutions

- Architected a multi-agent simulation framework evaluating 26 LLM agents across 30 rounds with dual-replication protocols, simulating climate shocks, Loss-and-Damage pooling, and democratic voting.
- Ran diff-in-diff econometric models ($p < 0.05$) across 114 qualitative post-event LLM rationales ($\kappa = 0.72$), proving bad-reputation gossip causes strategic withdrawal when peer-sanctions are absent.
- Designed a risk-sharing Loss-and-Damage fund module achieving 76.8% shock coverage, coupled with an interactive HTML/JS browser visualizer for real-time agent decision and wealth telemetry.

BATNA — Bilateral Agent Trust & Negotiation Architecture

- Built a LangGraph multi-agent system: two tool-calling LLM agents negotiate 6-term contracts via a deterministic Nash Bargaining engine.
- Grounded agents in live data via Model Context Protocol (MCP) servers offering BLS/FRED benchmarks and BM25/RRF retrieval over USAspending.gov awards, with dynamic tool discovery.
- Built a Theory-of-Mind auditor that flags fabricated citations, validated against 9 engineered hallucination cases.
- Codified a contract-risk engine with thresholds cited to legal standards (31 U.S.C. §3902, UCC §2-309, BLS PPI).
- Shipped real-time streaming (Redis + FastAPI + React) with 122 pytest tests, strict mypy/ruff, and GitHub Actions CI.

NYC Taxi Demand Forecasting — Production MLOps System

- Built an Optuna-tuned XGBoost pipeline achieving 0.949 Val R^2 , tracked via MLflow Model Registry.
- Served predictions through FastAPI backed by a WAL-mode SQLite feature store with TTL pruning.
- Deployed KS-test drift monitoring, validated end-to-end via simulated streaming replay against the live API.
- Shipped with 36/36 tests passing, zero ruff/mypy errors, fully containerised in Docker.

Education

Institute of Business Administration — BS Economics and Mathematics

June 2026

Key Courses: Foundations of Data Science, Linear Algebra, Probability Theory, Optimization Techniques, Applied Econometrics, Statistical Machine Learning, Stochastic Processes, Game Theory